

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] DECEMBER 2025

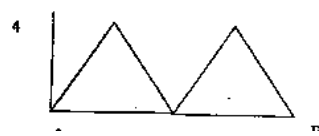
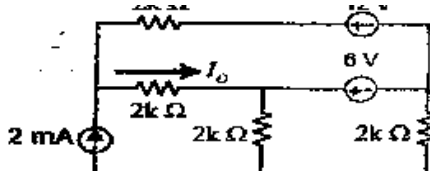
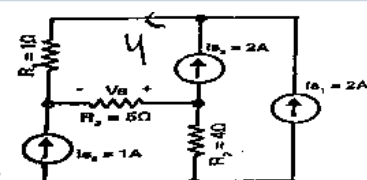
Paper Code: ES-107

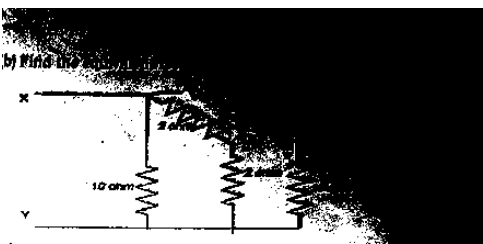
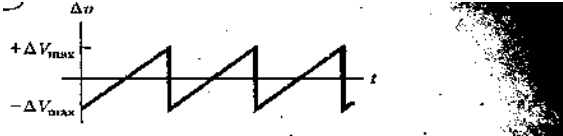
Subject: Electrical Science

Time: 3 Hours

Maximum Marks: 60

Note: Attempt all questions as directed. Internal choice is indicated.

Q. No.	Question	Marks
Q.1	Attempt any four of the following questions:	(4×5=20)
(a)	Define temperature coefficient of resistance. Explain the effect of temperature on resistance. Define resistivity.	5
(b)	Define Power factor. What are the different types of power in AC circuits?	5
(c)	For the waveform shown in Fig. 1 , calculate average value and RMS value.	5
		
(d)	Explain maximum power transfer theorem by considering resistive load in the circuit.	5
(e)	Explain the working principle of a Dynamometer type wattmeter.	5
(f)	What are the different types of three phase transformer connections? Explain star-star connection with its applications.	5
(g)	Write down the working principle and applications of an Auto Transformer.	5
(h)	A coil of resistance 8 Ω and inductance 0.12 H is connected in series across 230 V, 50 Hz supply. Calculate: (i) Impedance of the circuit (ii) Current flowing through circuit (iii) Power factor of the circuit (iv) Power drawn by the circuit	5
Q.2	Calculate current I₀ using superposition theorem in the network shown.	6
(a)		
(b)	State and explain Thevenin's theorem.	4
OR		
Q.3	Calculate node voltage V_a using Nodal analysis in the given circuit.	5
(a)		

Q. No.	Question	Marks
Q.3 (b)	Find the equivalent resistance between terminals x and y for the bridge/delta- Δ -wye circuit shown. 	5
Q.4 (a)	Find out the RMS value of the sawtooth waveform shown in figure. 	5
(b)	Explain resonance in series and parallel R-L-C circuits. Draw the resonance curve.	5
OR		
Q.5	Derive the formula to calculate Power in a three phase circuit.	10
Q.6 (a)	Explain different types of speed control methods for a DC shunt motor.	5
(b)	Derive the EMF equation for the generated voltage of a DC generator.	5
OR		
Q.7 (a)	What are the different types of AC three phase motors? Which type of motor is used for high starting torque?	5
(b)	Write down applications of a Synchronous motor. Give its merits.	5
Q.8 (a)	What are the different types of tests conducted on a transformer to find losses in the transformer?	5
(b)	Define Voltage regulation. Draw the phasor diagram of a transformer at inductive load.	5
OR		
Q.9 (a)	What are the different types of forces / torques acting on indicating instruments?	5
(b)	Explain the working principle of moving coil instruments. What are the advantages and disadvantages of moving coil instruments?	5

**** End of Question Paper ****